

Program:	Electrical Engineering	Course Code:	EE221
Duration:	180 Minutes	Semester:	Spring 2018
Paper Date:	29-05-2018	Total Marks:	100
Section:	ALL	Weight:	50%
Exam Type:	Final	Page(s):	5

Name: _____

Roll No: _____

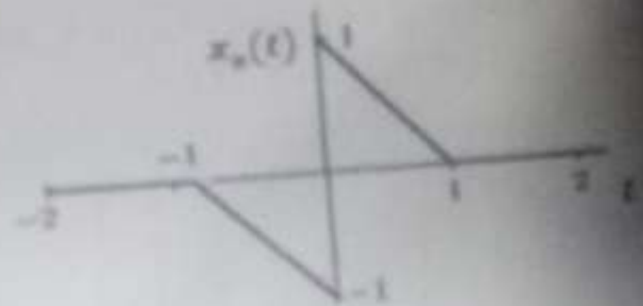
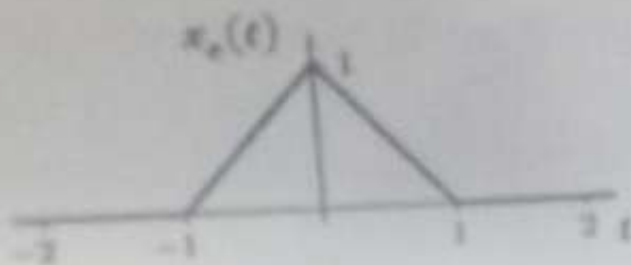
Section: _____

Notes: Carefully read the paper and attempt each question (only once). Attempt all parts of a question together.

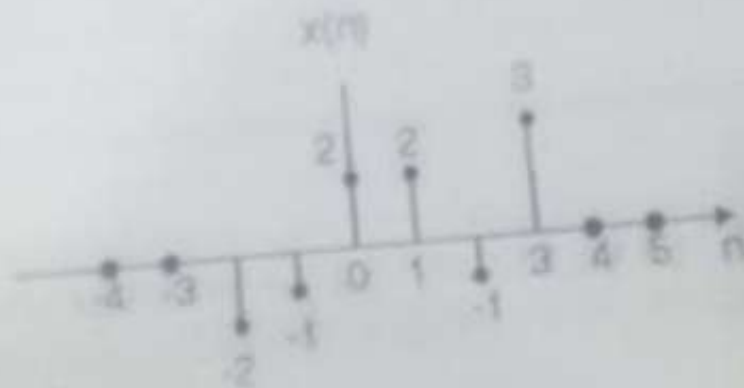
Q 1 (CLO # 1)

20 Marks

Even and odd components of a continuous time signal $x(t)$ are given below. Sketch the original signal $x(t)$.



Q 2 Determine the power and energy of following signal and state with reason if it is power signal or energy signal.



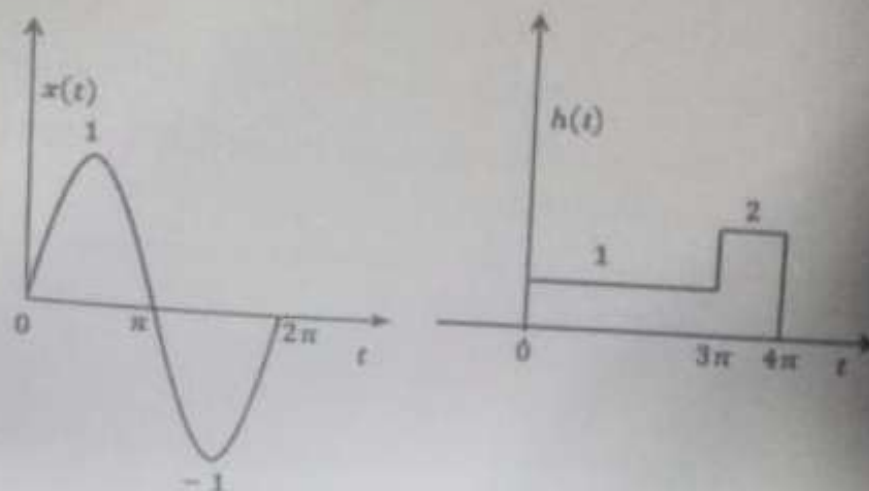
Q 3 Compute the following integral.

$$\int_{-\infty}^{+\infty} e^{j\omega t} \cos\left(\frac{\pi t}{3}\right) \delta(-t + 15) dt$$

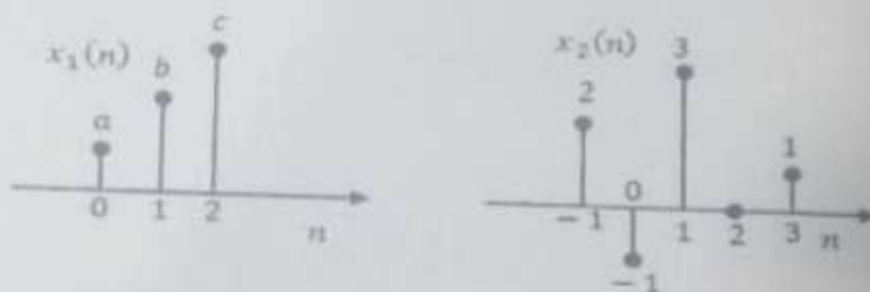
Question: 2 (CLO # 2)

20 M

- a) Two continuous time signals $x(t)$ and $h(t)$ are given below. Using graphical convolution techniques, find the mathematical expression for output such that $y(t) = x(t) * h(t)$.



- b) A discrete time signal $x_1(n)$ is convolved with another signal $x_2(n)$ to get the result $y(n) = x_1(n) * x_2(n)$.



- For what range of 'n', is the output non-zero?
- Find all non-zero sample values of $y(n)$.

Question: 3 (CLO # 7)

10 M

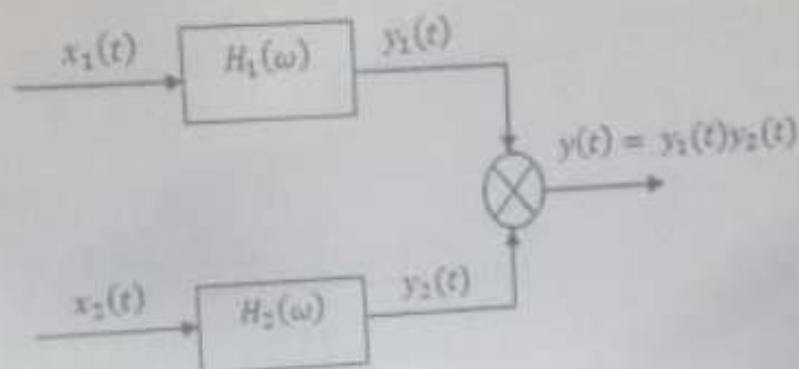
Find the z-transform of the following signal by definition (completely solve) and identify the region of convergence. (draw and label ROC graphically)

$$x(n) = \left(\frac{1}{7}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(-n-1)$$

Question: 4 (CLO # 4)

15 Marks

Signals $x_1(t) = 10^4 \text{rect}(10^4 t)$ and $x_2(t) = \delta(t)$ are applied at the inputs of the ideal low pass filters $H_1(\omega) = \text{rect}\left(\frac{\omega}{40,000\pi}\right)$ and $H_2(\omega) = \text{rect}\left(\frac{\omega}{20,000\pi}\right)$. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$.



- I. Sketch $X_1(\omega)$ and $X_2(\omega)$.
- II. Sketch $H_1(\omega)$ and $H_2(\omega)$.
- III. Sketch $Y_1(\omega)$ and $Y_2(\omega)$.
- IV. Find the bandwidth of $y_1(t)$, $y_2(t)$, and $y(t)$.

15 Marks

Question: 5 (CLO # 6)

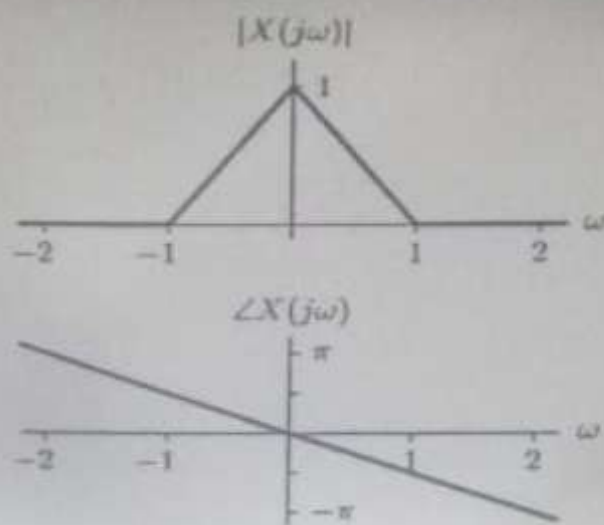
A signal $x(t) = \sin(5\pi t) + \cos(40\pi t)$ is sampled at a rate of 10 Hz.

- a) Find the simplified mathematical expression for the corresponding sampled signal $x(n)$.
- b) Sketch the discrete time signal found in part (a).
- c) Can $x(t)$ be reconstructed by low pass filtering the sampled signal $x(n)$? State your reason clearly.
- d) If we repeat the sampling of above signal $x(t)$ with a frequency of 40 samples per second, is it possible to reconstruct the signal properly? Explain with reason.
- e) What is the minimum sampling frequency required for proper reconstruction of signal $x(t)$?

Question: 6 (CLO # 5, PLO#1)

20

The magnitude and phase spectrum of a signal $x(t)$ are given in the following plots



You are required to find the magnitude and phase spectrum of the following five signals i.e. $z_1(t)$, $z_2(t)$, $z_3(t)$, $z_4(t)$ and $z_5(t)$ which are derived from $x(t)$

- a) $z_1(t) = \frac{dx(t)}{dt}$
- b) $z_2(t) = x(t) * x(t)$
- c) $z_3(t) = x(t - \frac{\pi}{2})$
- d) $z_4(t) = x(2t)$
- e) $z_5(t) = x^2(t)$

How to solve:

Six magnitude plots(M1-M6) and six phase plots(A1-A6) are given in Table below.

Determine which of these plots is associated with each of the above signals and place the appropriate label (e.g. M7 & A4 for $z_5(t)$).

Also clearly explain the reason for your choice (mathematically as well as in words).

You will get NO CREDIT if you give no explanation.

Note: more than one derived signals can have same magnitude and angle.

Table 1 for Question 6

